

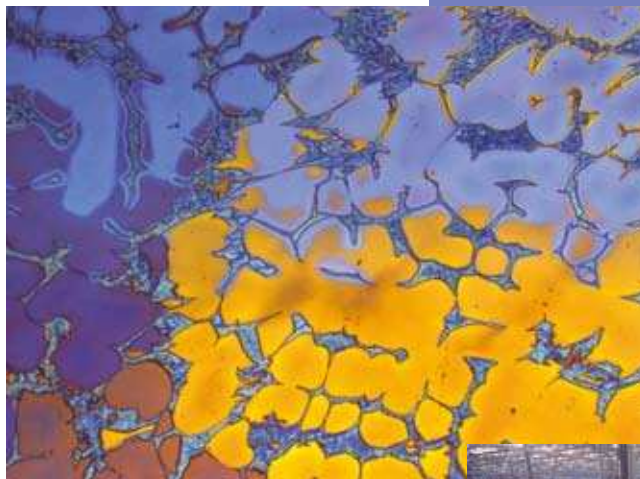
Metallographic preparation of copper and copper alloys

Copper, together with gold and tin were the first metals put to use by ancient man. The ease with which it could be shaped and formed, its attractive colour and its corrosion resistance were among some of the reasons why it found widespread use in the manufacture of early weaponry, jewellery, liturgical vessels and everyday domestic objects.

Metallic copper has been known since about 9.000 BC and was widely used by 5.000 BC. When its value as an alloy with tin was recognised, the birth of the Bronze Age initiated a profound cultural and economic development in Europe and the Mediterranean countries.

Historically, casting of bronze has a long tradition, but it is only since the 1920s that hot rolling and extrusion of copper and its alloys have been practiced. Nowadays, most of the copper produced is in wrought form, and with its high electrical conductivity copper is much used as cables, switch-gear components, transformers, motor windings, and generators. Oxygen free copper is mainly used in the electro-nics industry.

The corrosion resistance and high thermal conductivity of copper make it suitable for tubes, vessels and heat exchangers in the chemical, and food and beverage industry. Apart from the traditional applications of



Aluminium bronze, CuAl8, shows dendritic structure with α - δ eutectoid. Colour etching according to Klemm, pol. light, 100x.

copper for water and heating pipes, architects have in recent years discovered the appeal of designing facades with oxidised copper sheets.

The metallography of copper and its alloys is used for grain size measurement and purity checks by qualifying and quantifying the copper oxide content. On occasion, in certain brasses, the distribution of lead is determined as it can influence the machining process. In cast alloys the general microstructure and distribution of eutectic or lead is evaluated as well as the presence of shrinkage cavities or porosity.



Difficulties in the preparation of copper and its alloys

Pure copper is soft and ductile, easy to deform and prone to form scratches.

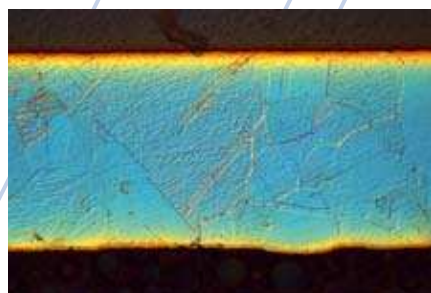


Fig. 1: Pure copper wire, with final polish of OP-S, DIC, 200x.

Bronzes, and even some of the harder brasses, may be susceptible to severe scratching.

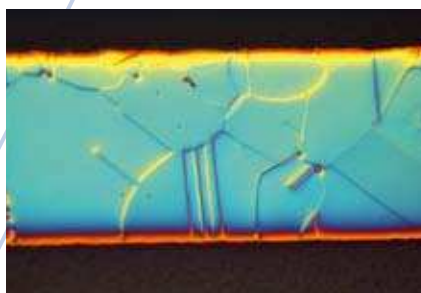


Fig. 2: Same sample as Fig. 1 with final polish of OP-S-ammonia water-hydrogen peroxide-mixture, DIC, 200x.

Solution

- Avoiding coarse grinding abrasives
- Thorough diamond polishing with soft cloths
- Chemical-mechanical fine polishing

Production and application



Fig. 3:
Electrolytic refining

Copper
from raw material to electrolytic copper

Although metallic copper occurs naturally, it is mainly extracted from sulphide ores in a metallurgical smelting process. A small amount of copper is also produced by hydro-metallurgical process. A brief description of the four steps in the copper production process is provided below.

1. Copper matte, 75% Cu:

In this first smelting process for extraction, copper concentrates, mainly chalcopyrite (CuFeS_2), are roasted and melted with fluxes in an oxygen flash furnace to matte copper. Matte is a mixture of copper sulphide and iron sulphide and contains approx. 75% copper.

2. Blister copper, 96-98% Cu:

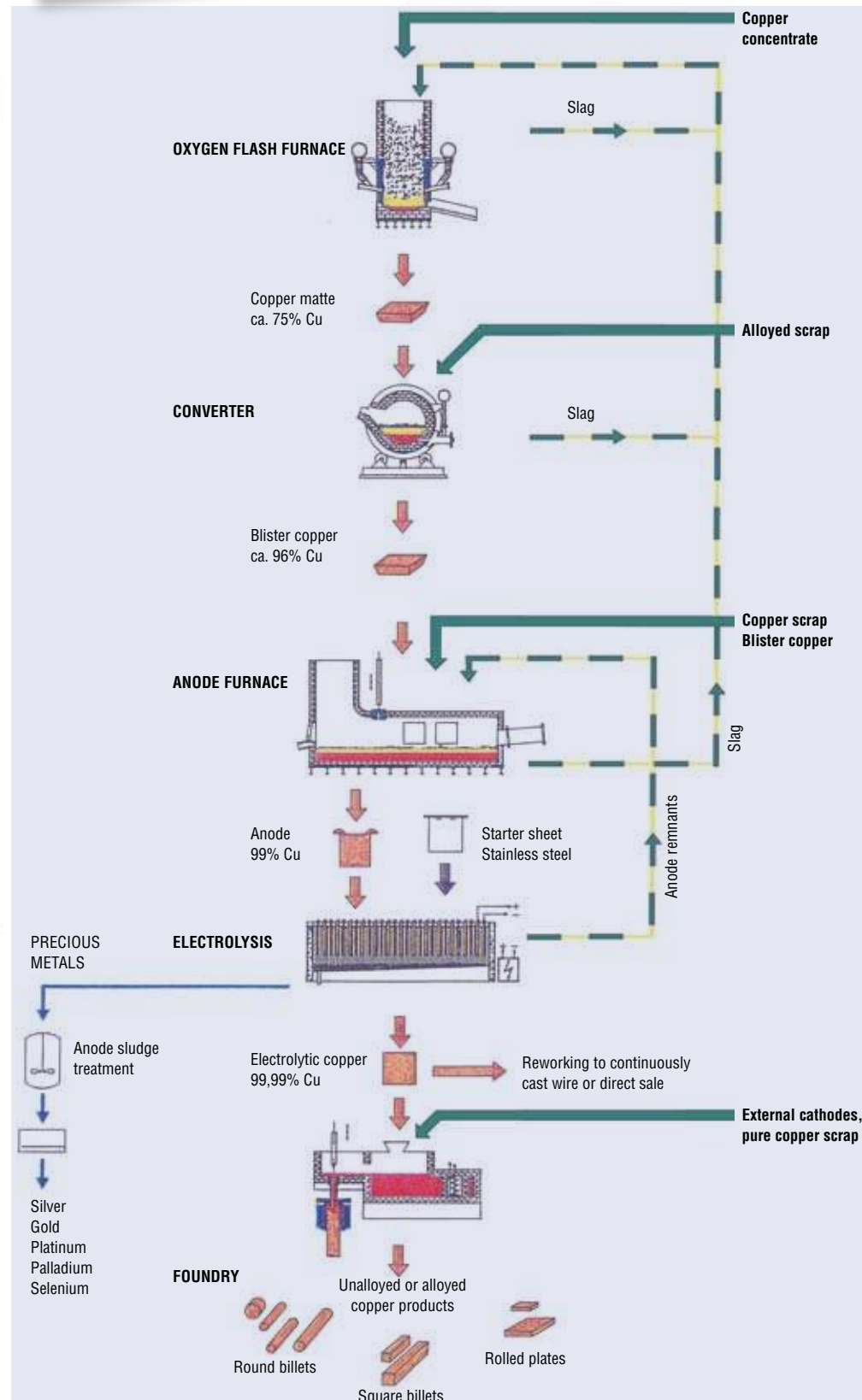
In a converter air is blown into the liquid matte to oxidise the sulphides. The resulting blister copper contains approx. 96-98% copper.

3. Anode copper, 99% Cu:

Blister copper is refined in the anode furnace. It is melted with copper scrap and rests of anodes from the electrolytic refining. Air blowing the melt, produces an oxidizing atmosphere which reduces the impurities that are then discharged as slag. The high oxygen content of the melt needs to be lowered to less than 0.1% as copper oxides make the copper brittle. The oxygen content is lowered by blowing natural gas into the melt, which reduces the copper oxide to copper, releasing steam and carbon dioxide. The resulting copper is 99% pure and cast into anodes that are subsequently used for electrolytic refining.

4. Electrolytic refining, copper cathodes, 99.99% Cu:

The copper anodes are still contaminated with metals such as Ni, Pb, Ag, Pd and Au. During the electrolytic refining process, high purity copper is produced whereby the impurities drop to the bottom of the electrolyses cell and are recovered. A solution of sulphuric acid and copper (II) sulphate, functions as electrolyte. Using direct current, the oxidation occurs at the anode, which is dissolved, and the pure copper is deposited at the cathode "starter sheet" of



Simplified schematic diagram of
copper production



Fig. 4:
Control valve for the distribution
of potable water.



Fig. 5: Copper with red copper oxides,
dark field, 500x.

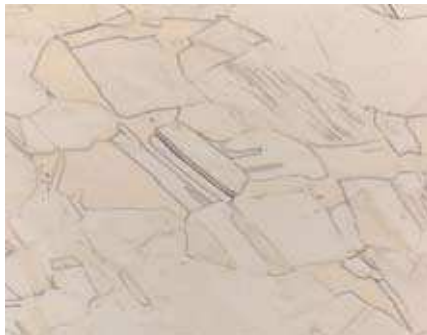


Fig. 6: Oxygen free copper, etched with ammonium
peroxydisulfate, 100x.



Fig. 7: Copper cathode, etched according
to Klemm, 100x.

pure copper or stainless steel, from which
it is mechanically removed.

The cathode plates of this very pure copper
are melted with 50% pure copper scrap
and cast into plates, round and square
billets which are then converted to sheets,
pipes, wire and casts. (Almost half the de-
mand of copper today is covered by copper
scrap and recycled copper.)

With increasing purity, the electrical and
thermal conductivity, as well as the price
of copper increase. Therefore the use of
pure copper depends on the application of
the product. Due to its excellent properties,
oxygen free copper for electronics (OFE) is
used as a basic material for semi conduc-
tors, switches and sealing rings in vacuum
technology and electron tubes.

The largest amount of copper is used in
construction for facades, roofs, potable
water pipes and heating installations, as
well as the electrical parts industry for
motor coils, generators and power supply
systems. Additional fields of applications
are cryogenics and air conditioning tech-
nology, chemical industry and beverage
and brewing technology.

Copper alloys

There are a large range of copper alloys;
however, zinc (brass) and tin (bronze) are
the most important alloying elements for
copper. Some of the most important alloys
and their applications will be briefly de-
scribed in the following section.

Brasses are copper alloys with 5 - 45%
zinc. Copper has a high solubility for zinc
and the alloys are very homogenous. Brass
with less than 28% zinc is called red brass
and is especially suitable for machining.
With increasing zinc content the reddish
copper colour changes to the yellow of
brass. Alloys up to 37% zinc consist of
 α - solid solution and are suitable for cold
forming. With increasing zinc content the
brass becomes harder and therefore easier
to machine. From 38% zinc, the alloys
display a two-phase α - β microstructure
which is suitable for hot forming. By add-
ing aluminium, manganese, iron, nickel
and tin, special types of brass, which have
specific chemical or mechanical proper-
ties, can be produced. Additions of small
amounts of lead improve the machinability
of brass.

Depending on the zinc content, different
brasses are used in various fields of ap-
plication: from watches and jewellery to
electro-technology (CuZn5), to springs,
screws, pins, die forged parts (CuZn30),
to armatures (CuZn40) and bearings
for sea water resistant pump housings
(CuZn10Sn2).



Fig. 8: Anti-friction and roller bearing brass cages



Danish Embassy, Berlin, clad with oxidised
copper plates.

Courtesy: Danish Foreign Ministry



Fig. 9:
Wrought bronze tubes
and profiles

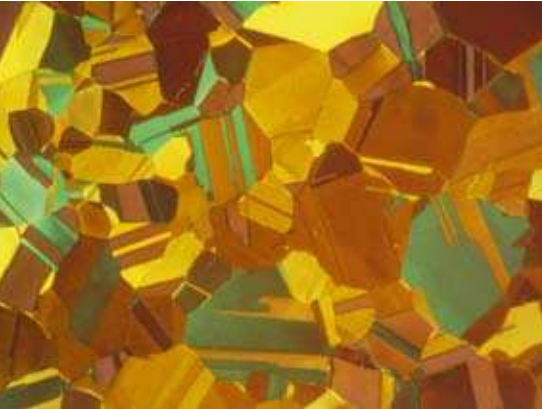


Fig. 10: α -brass, colour etched, 200x.

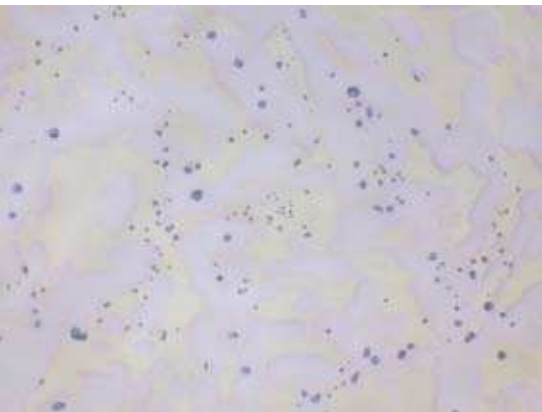


Fig. 11: α - β -brass cast (CuZn40Pb2) with grey-blue lead inclusions, unetched, 500x.

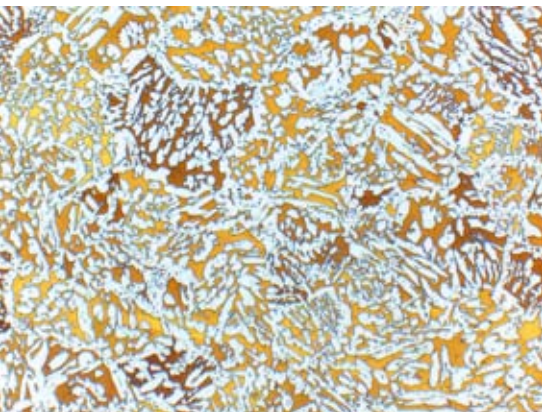


Fig. 12: α - β -brass cast, etched according to Klemm, light α -solid solution in dark matrix of β -solid solution, 100x.

Bronzes are copper-tin alloys that are divided into wrought alloys, with up to 8.5% tin, and cast alloys, with usually 9-12% tin, and also so-called bell casts with up to 20% tin.

Depending on the required properties, small amounts of zinc and phosphorus can be added to the wrought alloys, e.g. for bearings. These alloys are called phosphor bronzes. Lead, nickel and iron are common additions to cast alloys.

Cu-Sn-Zn-alloys are called gun metal. They are often used for the manufacture of slide bearings, worm gears, bearing bushes; parts which need a bearing surface capable of withstanding high contact loads. As well as good corrosion resistance, Gun metal has a low friction coefficient which means it can operate under such conditions without seizure.

Aluminium bronzes are copper alloys with up to 11% aluminium, displaying high strength at elevated temperatures and very good corrosion resistance. They are suited for marine propellers, highly stressed rotors for pumps and water turbines, and bearings and parts for the chemical industry. Copper-aluminium wrought alloys are used in mechanical and light engineering.

Beryllium bronze has high strength and hardness, with the specific property of not producing sparks when striking or impinging other metals. Therefore this type of bronze is suited for tools in an explosive environment such as refineries.

Due to their excellent corrosion resistance, copper-nickel alloys are used in desalination plants, for coins, heating wires in toasters and hair driers. The copper-nickel-zinc alloys, called German silver, display high strength, are corrosion resistant and are easily formed. Fields of application are surgical instruments, food industry, base metal for silver cutlery, contacts in connectors and jewellery.



Fig. 13: Cast bronze pump housing



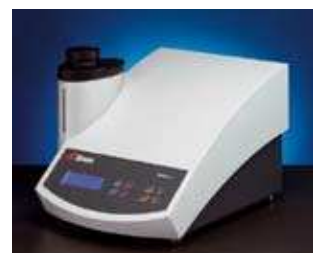
Fig. 14: Wrought bronze liners with graphite

Difficulties in the preparation of copper and copper alloys

With increasing purity, copper becomes softer and more susceptible to mechanical deformation and scratches. Consequently, grinding can cause deep deformation in high purity copper while grinding and polishing abrasives can be pressed into the surface. Copper alloys are harder, but still have a tendency to form scratches, which in some bronzes may only occur only in some individual grains.

Recommendations for the preparation of copper and copper alloys

For **sectioning** of copper a hard silicon carbide cut-off wheel is used that is generally suitable for non-ferrous metals. For **mounting**, a phenolic resin is in most cases sufficient.



CitoPress
Hot mounting press

Mechanical grinding and polishing

It is recommended that plane grinding is carried out with the finest possible grit in order to avoid any excessive mechanical deformation. Hardness, size and number of samples have to be taken into account, but even with larger samples of pure copper, plane grinding with 500# SiC-paper is sufficient. Large cast parts of copper alloys can be ground with 220# or 320#. It is also important that the force for grinding is low to avoid deep deformation.

As indicated in the table with the preparation data, for the soft alloys fine grinding with SiC-paper with fine grits is recommended, whereas MD-Largo with diamond can be used for the harder alloys. Better planeness and edge retention is obtained using MD-Largo.

Diamond polishing has to be carried out until all deformation and embedded abrasives from grinding have been removed. The chemical-mechanical fine polishing with silicon dioxide is especially important as an almost scratch free surface can be achieved. For pure copper, the final polish with a solution containing iron nitrate has proven to give very good results, and for copper alloys, a mixture of OP-S suspension with hydrogen peroxide and ammonia water is recommended (Formula below the table). After 1 minute polishing the result is checked under the microscope. If necessary the polish should be continued for another minute and the result checked again. It is recommended to continue this polish/check sequence until the required quality of the result has been achieved. If the attack is too fast or too strong the mixture should be diluted with water.

(Approx. 30 sec before the end of polishing, water is poured onto the polishing cloth to rinse the sample as well as the cloth. Then the sample is washed again with clean water from the tap and then dried.)

The preparation data are for the automatic grinding and polishing of 6 mounted samples, 30 mm diameter, clamped into a holder.

Electrolytic Polishing is suited for pure copper and α - brass wrought alloys. Two-phase α - β - brass can also be polished

Mechanical Preparation

Pure copper and copper alloys with low alloying contents.

Grinding

Step	PG	FG 1	FG 2	FG 3
Surface	SiC- Paper	SiC-Paper	SiC-Paper	SiC-Paper
Grit	320	800	1200	4000
Lubricant	Water	Water	Water	Water
rpm	300	300	300	300
Force (N)	150	150	150	150
Time	As needed	1 Min.	1 min.	1 min.

Polishing

Step	DP	OP	*Wet the polishing cloth with water and polish with a few drops of the etchant without any polishing medium. Formula: see table on page 6.
Surface	MD-Mol	OP-Chem	
Suspension	DiaPro Mol	Iron (III) nitrate*	
rpm	150	150	
Force (N)	150	90	
Time	4 min.	1 min.	

Copper alloys

Grinding

Step	PG	FG
Surface	SiC-Paper	MD-Largo
Körnung* / Suspension	220 or 320	DiaPro Allegro/Largo
Lubricant	Water	
rpm	300	150
Force (N)	180	180
Time	As needed	4 min.

Polishing

Step	DP	OP
Surface	MD-Mol*	OP-Chem
Suspension	DiaPro Mol	OP-S**
rpm	150	150
Force (N)	150	90
Time	3 min.	1-2 min.

*Alternative: MD-Dac

**96 ml OP-S, 2 ml ammonia water (25%), 2 ml hydrogen peroxide (3%)

electrolytically, but the results are not very well suited for quantitative analysis, especially if the alloy contains lead.

Prerequisite for a good electrolytic polish is a preceding fine grinding with SiC-paper up to 2400# or 4000#. Due to the different phases in cast alloys, they are not suited for electrolytic polishing.

Electrolyte:	D2
Area:	0.5 cm ²
Voltage:	24 V
Flow rate:	10
Time	20 sec

Directly after polishing the sample can be etched with the same electrolyte for 4 seconds with 2-4 volts.



Fig. 15: α - β -brass cast, mechanically polished, unetched, 200x.

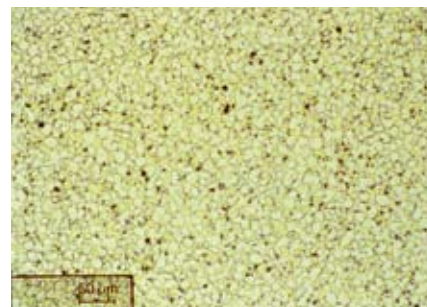


Fig. 16: Same specimen as Fig. 15 electrolytically polished, unetched, 200x. The lead inclusions are pulled out and therefore appear larger and more numerous than in Fig. 15.

Etching

There are numerous etchants for copper and its alloys that are relatively easy to apply. Most of the cast alloys are not difficult to etch. It can be more difficult to find the right etching solution for some of the wrought alloys, especially when they are severely cold worked. In these cases a colour etch can be helpful.

It needs to be pointed out that lead is attacked by the etchants and mostly only black voids remain. Micrographs that are to document the quantity and distribution of lead always have to be taken before etching. The colour of pure lead is grey-blue.

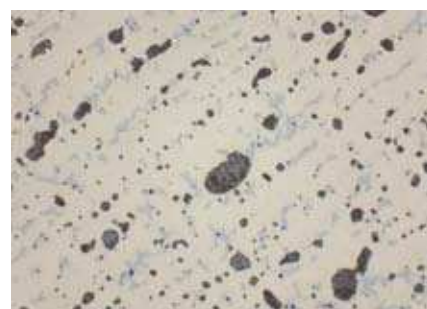


Fig. 17: Bronze cast, CuSn8Pb, unetched, large and small blue-grey lead inclusions, pale blue α - δ eutectoid discernable, 500x.

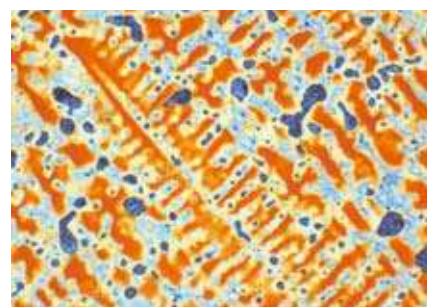
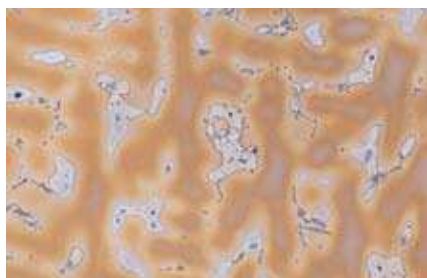


Fig. 18: Same specimen as in Fig. 17, colour etched according to Klemm, dendritic structure with light blue eutectoid and blue lead inclusions. Small lead inclusions can not be clearly differentiated, 500x.



Bronze cast, CuSn10, etched with iron (III) chloride, dendritic structure α - β eutectoid, 200x.

Application	Etchant
Grain area etch for copper, brass and bronzes	100 ml water 10 g ammonium peroxydisulfate Use fresh!
All types of copper	100-120 ml water or ethanol 20-50 ml hydrochloric acid 5-10 g iron (III) chloride (concentration variable)
Grain boundaries Grain areas	25 ml distilled water 25 ml ammonia water 5-25 ml hydrogen peroxide, 3%
α - β brass	120 ml Water 10 g copper (II) ammonium chloride Add ammonia water until precipitate dissolves
Fast and good polish for pure copper	100 ml water 100 ml ethanol 19 g iron (III) nitrate
Colour etch according to Klemm	100 ml cold saturated sodium thiosulfate 40 g potassium metabisulfite

Summary

Due to its good formability, high electric and thermal conductivity and corrosion resistance, pure copper is mainly used in electric engineering, the electronics industry and in the food and beverage industry. The application of brass and bronze wrought and cast alloys ranges from small parts to sea water resistant pump housings. The metallography of copper and its alloys is used in quality control mainly for checking purity and grain size determination. In addition, cast alloys are examined for general structure evaluation. Copper is soft and ductile and particularly prone to mechanical deformation. Therefore care needs to be taken that for the first grinding step the finest possible grit is used. Diamond polishing is carried out on medium soft to soft cloths and can take relatively long for pure copper. Chemical-mechanical final polishing with OP-S suspension is essential and results in a scratch free surface. Lead free wrought alloys can also be electrolytically polished. Etching with the common etchants is fairly easy, whereby colour etching can reveal some attractive structures in cast bronzes.

Authors

Elisabeth Weidmann, Anne Guesnier, Struers A/S

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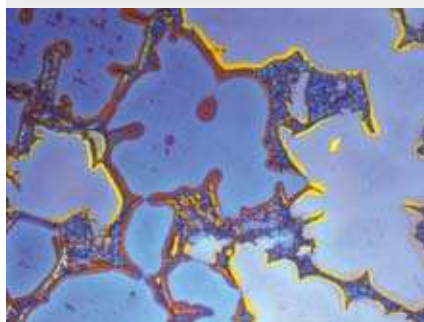
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Aluminium bronze, colour etched according to Klemm, pol. light, 200x.

USA and CANADA

Struers Inc.
24766 Detroit Road
Westlake, OH 44145-1598
Phone +1 440 871 0071
Fax +1 440 871 8188
info@struers.com

SWEDEN

Struers A/S
Smältvägen 1
P.O. Box 11085
SE-161 11 Bromma
Telefon +46 (0)8 447 53 90
Telefax +46 (0)8 447 53 99
info@struers.dk

FRANCE

Struers S.A.S.
370, rue du Marché Rollay
F- 94507 Champigny
sur Marne Cedex
Téléphone +33 1 5509 1430
Télécopie +33 1 5509 1449
struers@struers.fr

NETHERLAND/BELGIE

Struers GmbH Nederland
Electraweg 5
NL-3144 CB Maassluis
Tel. +31 (0) 10 599 72 09
Fax +31 (0) 10 599 72 01
glen.van.vugt@struers.de

BELGIQUE (Wallonie)

Struers S.A.S.
370, rue du Marché Rollay
F- 94507 Champigny
sur Marne Cedex
Téléphone +33 1 5509 1430
Télécopie +33 1 5509 1449
struers@struers.fr

UNITED KINGDOM

Struers Ltd.
Unit 25a
Monkspath Business Park
Solihull
B90 4NZ
Phone +44 0121 745 8200
Fax +44 0121 733 6450
info@struers.co.uk

JAPAN

Marumoto Struers K.K.
Takara 3rd Building
18-6, Higashi Ueno 1-chome
Taito-ku, Tokyo 110-0015
Phone +81 3 5688 2914
Fax +81 3 5688 2927
struers@struers.co.jp

CHINA

Struers Ltd.
Office 702 Hi-Shanghai
No. 970 Dalian Road
Shanghai 200092, P.R. China
Phone +86 (21) 5228 8811
Fax +86 (21) 5228 8821
struers.cn@struers.dk

DEUTSCHLAND

Struers GmbH
Karl-Arnold-Strasse 13 B
D-47877 Willich
Telefon +49(0)2154) 486-0
Telefax +49(0)2154) 486-222
verkauf.struers@struers.de

ÖSTERREICH

Struers GmbH
Zweigniederlassung Österreich
Ginzkeyplatz 10
A-5020 Salzburg
Telefon +43 662 625 711
Telefax +43 662 625 711 78
stefan.lintschinger@struers.de

SCHWEIZ

Struers GmbH
Zweigniederlassung Schweiz
Weissenbrunnstrasse 41
CH-8903 Birnsdorf
Telefon +41 44 777 63 07
Telefax +41 44 777 63 09
rudolf.weber@struers.de

CZECH REPUBLIC

Struers GmbH
Organizační složka
Havlíčková 361
CZ-252 63 Roztoky u Prahy
Tel: +420 233 312 625
Fax: +420 233 312 640
david.cernicky@struers.de

POLAND

Struers Sp. z o.o.
Oddział w Polsce
ul. Lirowa 27
PL-02-387 Warszawa
Tel. +48 22 824 52 80
Fax +48 22 882 06 43
grzegorz.uszynski@struers.de

HUNGARY

Struers GmbH
Magyarországi fióktelep
Puskás Tivadar u. 4
H-2040 Budaörs
Phone +36 (23) 428-742
Fax +36 (23) 428-741
zoltan.kiss@struers.de

SINGAPORE

Struers A/S
627A Aljunied Road,
#07-08 BizTech Centre
Singapore Post Centre
Singapore 389942
Phone +65 6299 2268
Fax +65 6299 2661
struers.sg@struers.dk